LINE BALANCING

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Line Balancing in Production Systems

- Line balancing is the process of assigning tasks to workstations in a production line so that the total time required at each workstation is approximately the same.
- It is a method of distributing workload evenly across workstations to optimize production flow, minimize idle time, and improve efficiency. It aims to synchronize operator and machine time to match the required production rate.

What is Line Balancing?

- - Process of assigning tasks to workstations
- - Goals:

1. Equalize Workloads
2. Minimizes idle time
3. Maximizes efficiency
4. Reduce cost
5. Improve productivity and efficiency
6. Identify and eliminate bottleneck
7. Determine production speed and manpower needed

Line Balancing

- -Improves productivity

Productivity refers to the efficiency with which resources are used to produce goods or services. It's a measure of how much output is achieved with a given amount of input, such as labor, capital, or materials. Productivity is a key performance indicator.

- - Reduces cycle time and bottlenecks.

Cycle time is the time it takes one workstation in a process to complete its workload for processing a part.
• Cycle Time is the total elapsed time to move a unit of work from the beginning to the end of a physical process.

- - Optimizes manpower and resource utilization

Examples of Unbalanced Lines

- - Station 1 overworked, Station 2 idle



- - Illustration: Car assembly with delays at paint station

Assembly Line

- - Single-model, Mixed-model lines

A single-model assembly line is a production system where a line is dedicated to assembling only one type of product. This contrasts with mixed-model lines, which assemble multiple product types. Single-model lines are typically used for high-volume production of a homogenous product.

- - Used in manufacturing, packaging, etc.



PC: Marek Ślusarczyk

https://upload.wikimedia.org/wikipedia/commons/9/9f/002_Production_line_-_car_assembly_line_in_General_Motors_Manufacturing_Poland_-_Gliwice%2C_Poland.jpg

Terminology

- - Workstation:

A specific area or location within a production line where a particular task or set of tasks is performed.

Example: Station for painting

- Task time: Task time refers to the time required to complete a specific operation or task at a workstation
- Cycle time: Cycle time refers to the total time it takes for a product to move through a specific stage or workstation in a production line. It essentially measures the time it takes for each unit of work to move from the beginning to the end of a particular process
- Precedence constraints: “Precedence constraints" refer to the required sequence of tasks or work elements that must be followed when assigning tasks to workstations. These constraints ensure that certain tasks cannot be completed before others, impacting the order of operations along the production line.
- Line efficiency and Balance delay

Line Efficiency

- Line efficiency is a metric used to assess the effectiveness of line balancing, measuring how well the line is operating compared to its theoretical capacity.
- Balance Delay: It refers to the percentage of time a production line is idle due to uneven workload distribution among workstations or operators. It's a measure of how much wasted time exists on the line because some stations might be idle while others are overloaded

Assumptions in Simple Line Balancing Models

- - Tasks are predetermined and fixed
- - Task times are deterministic and known
- - No worker learning or fatigue considered
- Precedence relation is known
- Each task is considered indivisible and is to be performed at a single workstation
- **No Work-in-Progress (WIP) Buffer:** There is no buffering or storage between workstations.
- **No Parallel Stations:** All workstations are arranged in series, and no parallel processing is allowed.

Cycle Time & Takt Time

- - Cycle time is the actual time it takes to produce one unit, from start to finish, including all value-added and non-value-added activities
- -Takt Time = Available Time /Customer Demand

Line Efficiency & Balance Delay

- - Efficiency = Total Task Time / (Number of Stations × Cycle Time)

Suppose a work contains 10 activities. Sum of activity durations is 120 minutes. Let us assume that the cycle time is 30 minutes and there are 5 stations. Then the line efficiency=120/(30×5)=0.8. Percentage efficiency=80%

- - Balance Delay = 1–Efficiency

In the given example, Balance Delay =1–0.8=0.2

Smoothness Index

- - Measures distribution of workload
- - Lower index = better balance

$$S_I = \sqrt{\sum_{i=1}^{K} \left(T_{\max} - T_i \right)^2}$$

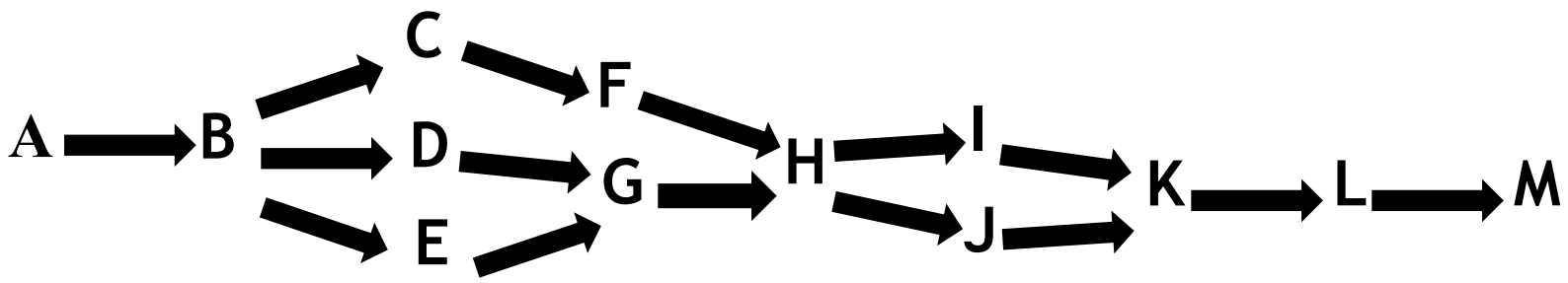
The value of *K* (number of workstations) can be between 1 to *N* (number of activities). T_i : Station time, T_{max} =max. station time

Steps in Line Balancing

- - 1. List tasks and times
- - 2. Identify precedence relationships
- - 3. Draw precedence diagram
- - 4. Select cycle time
- - 5. Assign tasks to workstations

Precedence Diagram

- - Graphical representation of task dependencies
- - Helps in identifying task sequences



Task Allocation Rules

- - Largest Candidate Rule (LCR): This rule prioritizes tasks with the longest durations when assigning them to workstations.
- Kilbridge and Wester: This method assigns tasks based on their position in the precedence diagram, grouping them into columns that reflect the sequence of operations. Tasks in earlier columns (i.e., those with fewer or no predecessors) are assigned to workstations first, moving progressively to later columns, while respecting precedence and cycle time constraints.
- - Ranked positional weight: This method considers both the task's duration and the durations of all subsequent tasks when determining its priority for assignment. Each task is given a "positional weight" that reflects its importance in the overall production flow. Tasks with higher positional weights are prioritized for assignment. A task that has a long duration and is followed by many other tasks will likely have a higher positional weight and be prioritized for assignment

Deciding number of stations

Number of stations required for designing a new station is estimated by takt time.
Say, a mobile phone company has the target to produce 400 units per day within an 6 hour shift.
The Takt Time is calculated by

$$\text{Takt Time} = \frac{\text{Available working time}}{\text{Customer Demand}}$$

$$= \frac{6 \times 60}{400} = 0.9\,\text{mins} = 54\,\text{s}$$

Each station should have at most 54 s cycle time to meet the demand of 400 units.

Say, the mobile phone company has to undergo 13 steps to complete the assembly of mobile phone. Steps are given by A–M.

Task	Time (s)
A	12
B	25
C	15
D	40
E	13
F	40
G	7
H	15
I	37
J	28
Kl	22
L	9
M	7
Total Time	**270**

Deciding number of stations

Number of stations is calculated by

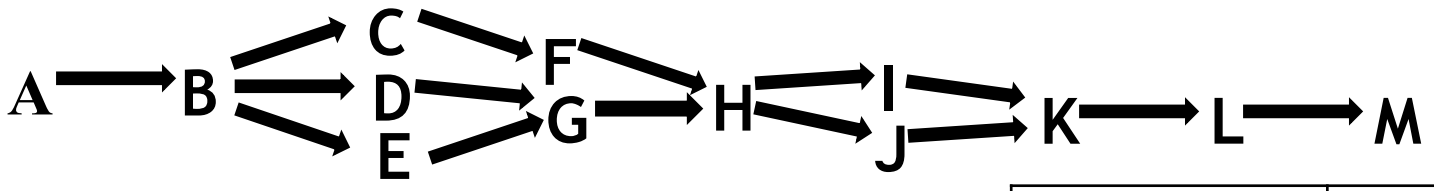
$$\text{Minimum Number of stations} = \frac{\text{Total task time}}{\text{Takt Time}}$$

$$= \frac{270}{54} = 5$$

13 tasks are to be completed within 5 stations. Some tasks have to be completed prior to other tasks. It has to be decided which tasks to include within a specific station. The order of carrying out the tasks are detailed by Precedence diagram.

Precedence diagram is a process flow diagram with arrows describing the steps within the assembly of the product or service.

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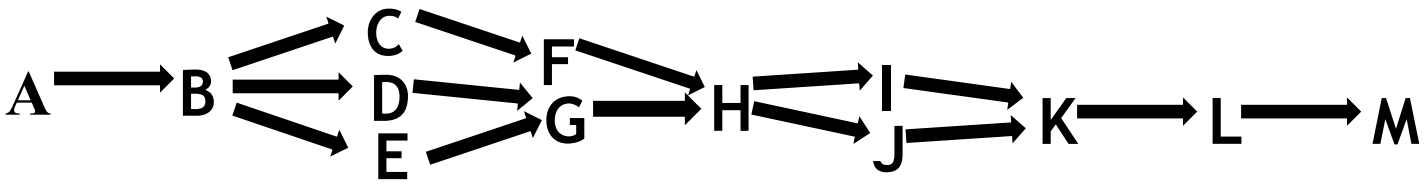


Precedence diagram comprising 13 tasks

It is inferred from precedence diagram that task B can be started only after the completion of task B. Tasks C, D and E can be started simultaneously after task B. Tasks F and G have to be completed before starting task H. Then, tasks I and J can be started simultaneously. After that, tasks K, L and M are to completed one after the other.

Next step involves assigning tasks to stations. An approach is to use "Task Assignment Table".

Task	Time (s)	Immediate Predecessor
A	12	--
B	25	A
C	15	B
D	40	B
E	13	B
F	40	C
G	7	D, E
H	15	F, G
I	37	H
J	28	H
K	22	I, J
L	9	K
M	7	L
Total Time	**270**	



Station	Eligible task	Assigned task	Task time (s)	Cumulative time (s)	Remaining time (s)
1	A	A	12	12	54–12=42
	B	B	25	12+25=37	54–37=17
	C, E	C	15	37+15=52	54–52=2
2	D, E	D	40	40	54–40=14
	E	E	13	40+13=53	54–53=1
3	F, G	F	40	40	54–40=14
	G	G	7	40+7=47	54–47=7
4	H	H	15	15	54–15=39
	I, J	I	37	15+37=52	54–52=2
5	J	J	28	28	54–28=26
	K	K	22	28+22=50	54–50=4
6	L	L	9	9	54–9=45
	M	M	7	9+7=16	54–16=38

Ranked Positional Weight Method

- - Calculate positional weights = task time + successors' time
- - Rank and assign tasks accordingly

Task	Immediate Predecessor(s)	Task Time (minutes)
A	—	5
B	A	3
C	A	2
D	B, C	4

Step 1: Find positional weights.

Positional weight of a task = its own task time + all successors' task times.

- **Task A** → 5 + 3 (B) + 2 (C) + 4 (D) = **14**
- **Task B** → 3 + 4 (D) = **7**
- **Task C** → 2 + 4 (D) = **6**
- **Task D** → 4 (no successors) = **4**

Step 2: Rank tasks by descending positional weight.

Order:

1. Task A (14)
2. Task B (7)
3. Task C (6)
4. Task D (4)

Step 3: Assign to workstations (if cycle time is given).

Parallel Workstations

- - Used when task time > cycle time
- - Improves throughput but increases cost

Mixed Model Assembly Lines

- - Different product variants on same line
- - Requires flexible scheduling and task planning

Two-sided Line Balancing

- - Used in car assembly lines
- - Tasks assigned to both sides of conveyor

Stochastic Line Balancing

- - Considers task time variability
- - Needs probabilistic models

Line Balancing in Industry

- - Automobile, Electronics, Appliance sectors
- - Case studies: Toyota, Dell

PC: Xu Hongzhi



https://commons.wikimedia.org/wiki/File:Jinbei_production_line.jpg

Software Tools for Line Balancing

- - Arena, Siemens Tecnomatix, Witness
- - Automate task assignments and simulation

Common Mistakes in Line Balancing

- - Ignoring precedence constraints
- - Overloading a station
- - Wrong cycle time selection

Summary of Key Concepts

- - Line balancing improves efficiency
- - Use appropriate heuristics or software
- - Evaluate line performance metrics

Reference:
https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://www.simul8.com/resources/an-example-of-line-balancing-with-simulation.pdf&ved=2ahUKEwj8tsOp0vWMAxXvSmwGHVVzAAMQFnoECEsQAQ&usg=AOvVaw0nGZ6LlbpJaOkFZ2XDiprR